

Unpacking the chain mediation process between transformational leadership and knowledge worker creative performance: evidence from China

Knowledge
worker
creative
performance

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Abstract

Purpose – This paper aims to draw upon motivated information processing theory to examine the sequential mediating roles of perspective taking and boundary spanning between transformational leadership and the creative performance of knowledge workers.

Design/methodology/approach – The study was carried out on a sample, including a dyad of 398 knowledge workers and their immediate supervisors in four research institutes in southwest China. The authors tested the theoretical model using structural equation modeling (SEM) and Mplus 7.0 software.

Findings – The results support the mediation model in which perspective taking was found to significantly and positively mediate the relationship between transformational leadership and boundary spanning. Boundary spanning was found to significantly and positively mediate the relationship between perspective taking and creative performance. Moreover, both perspective taking and boundary spanning were found to mediate the relationship between transformational leadership and creative performance.

Practical implications – The study findings imply that the transformational leadership behaviors of managers or supervisors nurture knowledge workers' perspective taking and their boundary spanning activities leading to creative performance.

Originality/value – The findings contribute new knowledge to the relationship between transformational leadership and creative performance by uncovering the causal chain of a cognitive mechanism (perspective taking) with a behavioral mechanism (boundary spanning).

Keywords Transformational leadership, Perspective taking, Boundary spanning, Creative performance

Paper type Research paper



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Introduction

Creative performance is the bringing out of unique and valuable ideas regarding processes, products, procedures and services by knowledge workers who serve as essential raw materials for development and possible implementation (Amabile, 2012). A knowledge worker is classified as an employee whose main capital or product is knowledge or a worker who “thinks for a living” (Davenport, 2005). The ability of knowledge workers to generate, develop and implement creative outcomes helps the organization respond to opportunities and threats, leading to innovation, adaptability and competitiveness (Shujahat *et al.*, 2019). One predictor of creative performance is transformational leadership – the visionary, charismatic and inspirational actions of a leader – which has been shown to motivate creative performance in organizations (Afsar *et al.*, 2017; Banks *et al.*, 2016; Li *et al.*, 2018a, 2018b).

Research on transformational leaders’ impact on creative performance has been on the increase since its inception (Li *et al.*, 2015; Ma and Jiang, 2018). Nonetheless, there is a need to understand the intervening processes that allow transformational leadership to evoke creative performance among knowledge workers (Kim and Shin, 2017; Le and Lei, 2017; Li *et al.*, 2018a, 2018b). This goes beyond the single or parallel mediators commonly the focus of research (Golden and Shriner, 2017; Jaiswal and Dhar, 2015; Wang *et al.*, 2016). For instance, Su *et al.* (2019) study on transformational leadership drew on the upper echelon theory to show collaborative management as a mediator that indirectly and positively affects individual innovation and is related to creative performance. Also, Dong *et al.* (2016) found that transformational leadership that focuses on individuals encourages skill development with indirect positive effects on individual creativity. Again, Mittal and Dhar (2015) used intrinsic motivational theory to show that creative self-efficacy mediates the relationship between transformational leadership and employee creativity.

Also, leadership is known to influence whether knowledge workers adopt creative cognitions and behaviors that generate creative outcomes (Helzer and Kim, 2019; Zhang and Bartol, 2010). For instance, Wang *et al.* (2014) used a social cognitive lens to establish that transformational leadership causes knowledge workers to have the self-efficacy needed to adopt creative role identities, which then improves creativity. However, today’s knowledge-based organizations tend to blur traditional boundaries so that collaborative creative processes are particularly salient (Su *et al.*, 2019).

Transformational leadership has been shown to intrinsically and extrinsically motivate creativity among followers (Chen *et al.*, 2016; Liu *et al.*, 2016; Ma and Jiang, 2018; Pan *et al.*, 2017; Zhang *et al.*, 2015). Likewise, a deep understanding of tasks requires employees to exchange information and integrate both parties’ interests and perspectives, which is known to be positively related to intrinsic motivation and creativity (Grant and Berry, 2011). But few studies have combined those observations to examine whether transformational leadership evokes such motivated information-processes that then lead to creative performance. Transformational leaders are also noted to know how to encourage and motivate employees to perform in ways that create meaningful change (Chen *et al.*, 2016; Kim and Shin, 2017). Consequently, this results in an engaged workforce that’s empowered to create, innovate and help shape an organization’s future success. Hence, the primary objective of this research is to investigate *how* transformational leadership shapes knowledge workers’ creative performance. Motivated information processing provides the underlying theory to explain how knowledge workers access, exchange and integrate information and knowledge to pursue creative performance (De Dreu *et al.*, 2006; Super *et al.*, 2016).

Motivated information processing theory explains that individuals form perceptions, attitudes and consequent behaviors according to the information they receive from significant others, such as leaders and coworkers (De Dreu *et al.*, 2008). Thus, significant others greatly affect employees' readiness to learn, understand the task and discern regarding relevant or irrelevant organizational issues (De Dreu *et al.*, 2006; Grant and Berry, 2011). Therefore, the charismatic, visionary and inspirational nature of transformational leadership should encourage prosocial attitudes among knowledge workers (Chen *et al.*, 2016; Nijstad and De Dreu, 2012). This, in turn, would motivate them to understand and empathize with differing frameworks, take the perspectives of others, and be driven toward creative collaboration (Castañeda *et al.*, 2018). We argue that through perspective taking – the cognitive act of perceiving from someone else's point of view (Wang *et al.*, 2017), knowledge workers are motivated to pursue linkages and interactions with parties and divisions among various external perspectives (Roberts and Beamish, 2017) which in turn leads to creative performance.

The study makes the following contributions to existing literature. First, we build on and expand the literature by developing a theoretical proposition arguing that transformational leadership indirectly affects creative performance among knowledge workers via the intervening mechanisms of perspective taking and boundary spanning (Wang *et al.*, 2014). Second, our investigation of the chain mechanisms complements studies that have focused on single or parallel mediations between transformational leadership and creative performance (Kim and Shin, 2017; Li *et al.*, 2015). We further answer calls for identifying the dynamic and complex intervening mechanisms that determine leadership influences on creative performance (Tierney, 2008). Third, we extend understandings of motivated information processing theory in relation to creative performance to address a critical research question: how do transformational leaders motivate knowledge workers to interact collaboratively and creatively with colleagues (Li *et al.*, 2018a, 2018b)?

Theoretical background and hypotheses

Motivated information processing theory

Motivated information processing theory explains that creative performance requires individuals to exchange and process information (Liu *et al.*, 2016; Nijstad and De Dreu, 2012), but first, they must understand others' preferences, priorities and needs (De Dreu *et al.*, 2008). Thus, contextual factors such as leadership behavior would influence information sharing and in turn, the sharing of dispersed and critical information (Super *et al.*, 2016).

We argue that transformational leadership generates creative performance among knowledge workers by causing them to be motivated to act for the benefits of others. Such a form of motivated processes instills altruistic as well as cooperative goals to achieve collective outcomes in organizational endeavors (De Dreu *et al.*, 2006; Grant and Berry, 2011). Indeed, transformational leadership motivates followers to consider the perspectives of others (Gregory *et al.*, 2011). As a result, they span boundaries in the search for information. Perspective taking and boundary spanning then boost creative performance.

Our sequential mediation differs from prior arguments that transformational leadership affects creative performance through intrinsic motivation, psychological or cognitive processes (Liu *et al.*, 2016; Ma and Jiang, 2018; Pan *et al.*, 2017; Su *et al.*, 2019). Instead, we explain that transformational leadership inspires a kind of motivation that induces followers to consider the perspectives of others and consequently expand their boundaries (Wang *et al.*, 2017) (Figure 1).

The mediating role of perspective taking

Motivated information process theory promotes effective information processing. It explains that when knowledge workers take the perspective of others and span boundaries, they have increased positive affect, persistence, risk-taking and cognitive flexibility (Tippmann *et al.*, 2016; Super *et al.*, 2016). Transformational leaders are noted for encouraging positive work climates and cultures (Ng, 2017), which increases followers' motivations to consider others' viewpoints and ideas (Grant, 2012). We argue that positive work climates then encourage perspective taking. As perspective taking is reciprocal, when subordinates perceive that their leader cares about them, they feel similarly positive toward other organizational members (Wang *et al.*, 2017). As a result, followers are committed to team interests, which increases cooperation, communication, conflict resolution, harmonious relationships, mutual helping and the sharing of creative information (Castañeda *et al.*, 2018; Hoever *et al.*, 2012; Wang *et al.*, 2017). Hence, we propose:

H1. Perspective taking mediates the relationship between transformational leadership and boundary spanning.

The mediating role of boundary spanning

Based on motivated information process theory, we assert that when knowledge workers take others' perspectives and span boundaries, they become essential conduits between otherwise diverse and disconnected members for transferring information and facilitating creative performance. Indeed, access to different views, ideas and information is known to positively influence creative performance in organizations (Hsiao *et al.*, 2012). Similarly, knowledge exchange through boundary spanning is critical for achieving comprehensive and novel solutions to problems (Tippmann *et al.*, 2016). These exchanges correspondingly explain the fact that as leaders and members repeatedly solicit, develop and share ideas through boundary spanning, they increase the collaborations that binds them to bring out novel and useful ideas.

Also, by ensuring that individuals and teams have diverse backgrounds that cross-departmental and corporate boundaries, organizations can foster deep and effortful information processing to enhance creative performance (Wang *et al.*, 2017). Boundary-spanning encourages information processing for mediating conflicts, building competence and negotiating meaning across boundaries to facilitate the creation of novel and useful ideas (Barner-Rasmussen *et al.*, 2014). We conclude that boundary spanning directly shapes the creative performance of knowledge workers (Andersen and Kragh, 2015; Korschun, 2012) by creating ties that strongly bind internal parties (Schotter *et al.*, 2017) and that perspective taking indirectly shapes their creative performance, hence we hypothesize:

H2. Boundary spanning mediates the relationship between perspective taking and creative performance.

Transformational leadership, perspective taking, boundary spanning and creative performance

Motivated information processing theory explains that increased positive affect encourages risk-taking, persistence and cognitive flexibility in all aspects of creative performance (Grant and Berry, 2011). Our leadership→cognition→behavior→creative performance

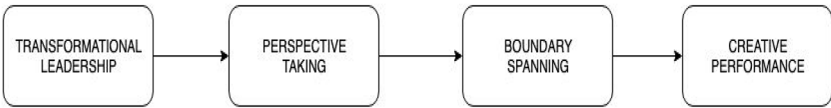


Figure 1.
Research model

framework illustrates that transformational leadership indirectly affects creative performance by granting knowledge workers significant responsibilities that allow them to choose novel ways to perform tasks (Groves, 2016; Li *et al.*, 2015). Personal control will enable them to influence others, encourage more creativity, share and solicit ideas among themselves and others conducting similar projects (Su *et al.*, 2019). Thus, transformational leaders enhance information processing climates and cultures through the integration of members interests that leads to perspective taking (Goldstein *et al.*, 2014) and encourages knowledge workers to exchange information and process information through boundary spanning (Le and Lei, 2017; Li *et al.*, 2018a, 2018b). In summary, transformational leaders directly change employees' perspectives, indirectly encourage boundary spanning and ultimately enhance creative performance. Hence, we hypothesize:

H3. Perspective taking and boundary spanning mediate the relationship between transformational leadership and creative performance.

Methods

Sample and procedure

Knowledge workers and their immediate leaders in four research institutes in the southwest of China were the participants in this study. The four research institutes were operating in the petrochemical, chemical, coal exploration, iron and steel industries. These members were encouraged to develop new products or improve the production process, which requires them to generate new and relevant ideas. To prevent common method bias, data were collected in two waves. Wave 1 (T1) survey of knowledge workers form contained the predictor variables (i.e. perceived transformational leadership of leaders), the first stage mediator (i.e. their perspective taking) and control variables (i.e. their demographics). Wave 2 (T2) survey of knowledge workers contained the second stage mediator (i.e. their boundary-spanning). In contrast, the survey of leaders included their demographics and the creative performance of individual members. Data collection was done through the following procedure.

With the assistance of the human resource (HR) departments, a team of graduate students surveyed 600 randomly selected team members and their corresponding leaders. Respondents were told that the survey was for a research project to let them know that anonymity is guaranteed. Separate questionnaires were then administered to the identified members and leaders. A researcher-identification number was given to each questionnaire. This was to match the responses of members (T1 and T2) with their leaders (T2). To ensure confidentiality, questionnaires were put in sealed envelopes and returned directly to the research team. Each respondent received a token gift for participating in the survey.

The research team received completed responses from 528 members of 186 teams in the first wave, representing a response rate of 88%. Approximately three months later, the employees and their immediate leaders were contacted, and the second wave surveys were delivered to 178 leaders and 469 members from 178 teams. This period, 412 members (87.84% response rate) and 165 supervisors (92.69% response rate) returned questionnaires they had completed. After matching the leader-member responses at two-time points, we obtained 398 leader-member dyads, comprising 398 members and 159 leaders. For the member sample, 51.75% were male. For the leader sample, 79.75% were male. The average age of the members and leaders were 30.3 years old ($SD = 6.25$) and 37.06 years old ($SD = 6.13$), respectively. Regarding their education, 16.50% of members and 14.25% of leaders held less than a bachelor's degree, 52.00% of members and 50.00% of leaders had bachelor's degrees, and 31.50% of members and 35.75% of leaders held postgraduate's degrees. Analysis of variance (ANOVA) tests revealed no significant differences in gender, age and education between respondents and non-respondents

among leaders and members. The four research institutes did not show any significant differences among the samples about any demographic variable.

Measures

Entirely, the measures used in our survey were in English and was translated into Chinese, adopting Brislin's (1980) 'translation and back-translation' technique. To avoid central tendency bias among respondents the ratings were done using a six-point scale, ranging from strongly disagree (1) to strongly agree (6).

Transformational leadership. We used the sixteen-item measurement of the Multifactor Leadership Questionnaire (Bass and Avolio, 1997) to measure transformational leadership. This scale consists of four dimensions: idealized influence (e.g. "Our team leader emphasized the importance of having a collective sense of mission"), inspirational motivation (e.g. "Our team leader expresses confidence that goals will be achieved"), individualized consideration (e.g. "Our team leader helps others to develop their strengths") and intellectual stimulation (e.g. "Our team leader gets others to look at problems from many different angles"). To test a higher-order model, we performed a confirmatory factor analysis (CFA) to capture the variance in the four transformational leadership dimensions. The fit indices for the higher-order model fell within a good range ($\chi^2 = 175.23$, $df = 89$, $p < 0.001$; RMSEA = 0.05, CFI = 0.98, IFI = 0.98, TLI = 0.96, SRMR = 0.04). All the four dimensions loaded significantly at the 0.001 level on the latent variable of transformational leadership. The errors were not allowed to covary. We compared the four-factor first-order model ($\chi^2 = 196.68$, $df = 87$, $p < 0.001$; RMSEA = 0.06, CFI = 0.96, IFI = 0.96, TLI = 0.94, SRMR = 0.04) with the higher-order model's fit. The CFA results demonstrated that the higher-order model yielded a similar fit to the four-factor first-order model. Because the research was based on the effect of transformational leadership as a whole construct on subordinate creative performance instead of differential influences of the various dimensions, previous studies were followed (Bai *et al.*, 2016) to combine the four factors into one transformational leadership factor. The internal reliability for this scale is 0.90.

Perspective taking. Perspective taking was accessed using the four-item subscale taken from Davis (1996). A sample item was, 'On the job, I frequently try to take other people's perspective. The internal reliability of this scale is 0.73.

Boundary spanning. Boundary spanning was accessed using the four-item subscale taken from Marrone *et al.* (2007). A sample item was, 'I proactively seek the advice and support of my faculty advisor'. The internal reliability of this scale is 0.84.

Creative performance. Creative performance was accessed using the four-item subscale taken from Farmer *et al.* (2003). A sample item was, 'This employee tries new ideas or methods first'. The internal reliability for this scale is 0.92.

Control variables. Both the member and leader's demographic variables such as age, gender and educational levels, as well as the member's tenure with the leader were statistically controlled for in our study because prior studies indicated that such variables are possibly associated with member creative performance (Gong *et al.*, 2009; Liao *et al.*, 2010; Ng and Feldman, 2013; Zhang and Bartol, 2010). Respondents were requested to self-report their age and tenure with their leaders in years. Gender was given these dummy codes; female (0) and males (1). Education was also given the following codes; employees who had high school education or lower were given (1), those that had a college education (2), those that had a bachelor's degree (3), and those that had a postgraduate degree (4).

Confirmatory factor analyses

We did run a series of confirmatory factor analyses (CFAs). This was to guarantee that there is convergent and discriminant validity within the four variables of the research. The

hypothesized four-factor model (i.e. transformational leadership, perspective taking, boundary spanning and creative performance) fit well into the data, through $\chi^2 = 767.66$, $df = 335$, $\chi^2/df = 2.29$, CFI = 0.92, IFI = 0.92, TLI = 0.91, RMSEA = 0.06, SRMR = 0.05. Also, there was convergent validity because all the factor loadings proved significant. We tested the discriminant validity of all four constructs through placing three alternative models in contrast to the four-factor model: In the first Model, transformational leadership and perspective taking were merged in one factor, boundary spanning and creative performance were divergent ($\chi^2 = 878.75$, $df = 338$, $\chi^2/df = 2.60$, CFI = 0.90, IFI = 0.90, TLI = 0.88, RMSEA = 0.06, SRMR = 0.06); in the second Model, in which transformational leadership, perspective taking and boundary spanning were merged into one factor and creative performance was divergent ($\chi^2 = 1169.89$, $df = 340$, $\chi^2/df = 3.44$, CFI = 0.84, IFI = 0.84, TLI = 0.82, RMSEA = 0.08, SRMR = 0.08) and the third Model, all four factors were merged into a single major factor ($\chi^2 = 1763.65$, $df = 341$, $\chi^2/df = 5.17$, CFI = 0.73, IFI = 0.70, TLI = 0.73, RMSEA = 0.10, SRMR = 0.11). The CFA results show that all the three Models produced poor fit indices. The chi-square differences show that M1: $\Delta\chi^2(3) = 11.09$, $p < 0.05$; M2: $\Delta\chi^2(5) = 402.23$, $p < 0.05$; and M3: $\Delta\chi^2(6) = 995.99$, $p < 0.05$. The discriminant and convergent validity of our measures were supported.

Data analyses

Structural equation modeling (SEM) with MPLUS 7.0 software was used to test the model proposed by the study. A confirmatory factor analysis was conducted to evaluate the measurement model and the two-step approach to SEM by [Anderson and Gerbing \(1988\)](#). A series of structural models followed this. Some goodness-of-fit indices in evaluating the fit of the model used by the study were reported.

The indices used included the chi-square statistic which was divided by the degrees of freedom (χ^2/df), the comparative fit index (CFI), the root mean square error of approximation (RMSEA), the Tucker–Lewis coefficient (TLI) and the standard root-mean-square residual (SRMR). In assessing the model fit, we used the goodness-of-fit indices proposed by [Joreskog and Sorbom \(1993\)](#) and Kline (1998) in the following manner: the χ^2/df ratio is suggested to be lower than three; the values of CFI and TLI are suggested to be more than or equal to 0.90; the RMSEA and SRMR are recommended to be less than 0.06. All these indices are parsimony-adjusted ([Maruyama, 1998](#); [Tanaka, 1993](#)) because each parameter estimated could be adjusted. A null reference model was used to calculate the goodness of fit of TLI and CFI. The constructs in the research are represented by more than three items.

Results

Descriptive statistics

[Table 1](#) presents the standard deviations, means, correlations between the research variables and reliabilities. The bivariate correlations point out the fact that transformational leadership is positively associated with perspective taking ($r = 0.19$, $p < 0.01$), boundary spanning ($r = 0.32$, $p < 0.01$) and creative performance ($r = 0.20$, $p < 0.01$); Perspective taking is positively related to boundary spanning ($r = 0.30$, $p < 0.01$) and creative performance ($r = 0.05$, $p < 0.05$). There is a positive correlation between boundary spanning and creative performance ($r = 0.17$, $p < 0.01$). The composite reliabilities and the variance for every construct in the study's model were first computed as posited by [Fornell and Larcker \(1981\)](#) and [Werts et al. \(1974\)](#). Values on the diagonal (in parentheses) represent Cronbach's alphas.

Table 1.
Means, standard
deviations and
correlations

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10	11
1. Supervisor age	37.06	6.13											
2. Supervisor gender	0.79	0.40	0.13 [*]										
3. Supervisor education	3.24	0.70	-0.22 ^{**}	0.11 [*]									
4. Subordinate age	30.34	6.25	0.34 ^{**}	-0.02 [*]	-0.06								
5. Subordinate gender	0.52	0.50	0.02	0.12 [*]	0.11 [*]	0.13 ^{**}							
6. Subordinate education	3.13	0.73	-0.13 ^{**}	0.09	0.42 ^{**}	-0.26 ^{**}	0.19 ^{**}						
7. The subordinate's tenure with supervisor	3.40	2.69	0.32	-0.10 [*]	-0.20 ^{**}	0.57 ^{**}	-0.03	-0.27 ^{**}					
8. Transformational leadership (T1)	4.62	0.64	-0.01	0.01	0.03	-0.01	-0.05	-0.04	-0.09	(0.90)			
9. Perspective taking (T1)	4.55	0.53	-0.00	-0.02	0.85	-0.06	0.02	0.05	-0.02	0.19 ^{**}	(0.73)		
10. Boundary spanning (T2)	4.47	0.76	0.09	0.07	0.00	0.06	0.06	-0.10	0.09	0.32 ^{**}	0.30 ^{**}	(0.84)	
11. Creative performance (T2)	4.58	0.91	0.08	0.10 [*]	0.11 [*]	0.07	0.10 [*]	0.08	0.06	0.22 ^{**}	0.05 [*]	0.17 ^{**}	(0.93)

Notes: $n = 398$; ^{**} $p < 0.01$; ^{*} $p < 0.05$. Reliability (alpha) appears within parentheses on the diagonal

Model comparisons and hypothesis tests

The ensuing discussions present the results of the tested hypothesis, demonstrating the mediating relationships by a sequence of models (as depicted in Table 2). Table 2 presents the result that the baseline model fits the data perfectly ($\chi^2 = 971.33$, $df = 508$, CFI = 0.91, TLI = 0.90, RMSEA = 0.05, SRMR = 0.06). Apart from control variables, all other paths remain significant. The research also tested four related models. The first Model is similar to the baseline model, only that perspective taking and boundary spanning is reversed ($\chi^2 = 972.82$, $df = 508$, CFI = 0.91, TLI = 0.90, RMSEA = 0.05, SRMR = 0.06). The second model is in close relation with the baseline model, only that direct connection from transformational leadership to boundary spanning is added ($\chi^2 = 951.04$, $df = 507$, CFI = 0.92, TLI = 0.91, RMSEA = 0.05, SRMR = 0.06). The third Model is similar to the baseline model, with the exception that straightforward pathways from perspective taking to creative performance are added ($\chi^2 = 971.03$, $df = 507$, CFI = 0.91, TLI = 0.90, RMSEA = 0.05, SRMR = 0.06). The fourth Model is closely related to the baseline model, except that the direct pathways from transformational leadership to boundary spanning and also from perspective taking to creative performance are added ($\chi^2 = 952.11$, $df = 506$, CFI = 0.92, TLI = 0.91, RMSEA = 0.05, SRMR = 0.06).

None of the added pathways significantly improved the fit of any of the models, even though all the four models have a greater number of pathways than the baseline mediator model. Some additional paths were not statistically significant. For instance, the path in Model 1 from perspective taking to creativity is not significant. However, the model fit of Model 1 is as good as the baseline model. This makes the baseline model more fitting for the study. Table 2 shows the model of the study and its standardized coefficients.

The model and its standardized coefficients are depicted in Table 2. A positive association was found between transformational leadership and perspective taking ($\beta = 0.15$, $p < 0.01$); there is a positive and significant association between perspective taking and boundary spanning ($\beta = 0.58$, $p < 0.001$), and boundary spanning also has a significant and positive association with creative performance ($\beta = 0.24$, $p < 0.01$). The indirect impact of transformational leadership and boundary spanning on perspective taking was found to be significant ($\beta = 0.08$, $p < 0.01$), perspective taking and creative performance through boundary spanning was found to be significant ($\beta = 0.14$, $p < 0.01$) and transformational leadership and creative performance through perspective taking and boundary spanning was also found to be significant ($\beta = 0.02$, $p < 0.05$). The findings are therefore in support of the model hypothesized by the study and *H1* to *H3* as well.

Discussion

The objective of this study was to integrate transformational leadership, perspective taking and boundary spanning to understand how they interact in the motivated informational process that leads to creative performance. The results support our postulation that transformational leaders encourage employees to consider others' viewpoints about organizational activities (Gregory *et al.*, 2011). Perspective taking then significantly and positively motivates boundary spanning, mediating the link between transformational leadership and boundary spanning. Furthermore, boundary spanning is positively related to creative performance, as previously shown by Andersen and Kragh (2015), while mediating the relationship between perspective taking and creative performance. Finally, perspective taking and boundary spanning mediate the relationship between transformational leadership and creative performance. The findings extend understandings of motivated information processing mechanisms and conditions through which transformational leadership impacts knowledge workers (De Dreu *et al.*, 2006). More importantly, the results reveal that transformational leaders evoke positive motivational states' that transform

Table 2.
Unstandardized path
coefficients and
indirect effects in
models

Model	Model	Model	Model	Model	Model
	Path coefficients/ fit indices	Model	Path coefficients/ fit indices	Model	Path coefficients/ fit indices
Baseline model					
TFL → PT	0.15**	TFL → BS	0.40***	TFL → PT	0.14**
PT → BS	0.58***	BS → PT	0.22***	PT → BS	0.59***
BS → CT	0.24**	PT → CT	0.16	BS → CT	0.26**
				TFL → PT	0.34***
				PT → CT	−0.09
TFL → PT → BS	0.08**	TFL → BS → PT	0.09***	TFL → PT → BS	0.06**
PT → BS → CT	0.14**	BS → PT → CT	0.04	PT → BS → CT	0.11**
TFL → PT → BS → CT	0.02*	TFL → BS → PT → CT	0.01	TFL → PT → CT	−0.01
				TFL → BS → CT	0.09**
				TFL → PT → BS → CT	0.02*
χ^2	971.33	χ^2	972.82	χ^2	952.11
df	508	df	508	df	506
CFI	0.91	CFI	0.91	CFI	0.92
TLI	0.90	TLI	0.90	TLI	0.91
RMSEA	0.05	RMSEA	0.05	RMSEA	0.05
SRMR	0.06	SRMR	0.06	SRMR	0.06

Notes: $n = 398$; ** $p < 0.01$; * $p < 0.05$. TFL = Transformational Leadership; PT = Perspective Taking; BS = Boundary Spanning; CRE = Creative Performance. Each model also tested the control variables of subordinate and supervisor's age, gender and educational levels, as well as the subordinate's tenure with the supervisor as exogenous antecedents of subordinate creativity

knowledge workers' perspectives and information exchange processes (Goldstein *et al.*, 2014). The interactions then motivate workers to produce novel and useful ideas.

Theoretical implication

First, an overall contribution to the transformational leadership literature is finding a new mechanism behind the relationship between transformational leadership and creativity by highlighting the mediating role of perspective taking and boundary spanning in such a relationship. Meta-analytic research suggests the need to reconcile inconsistent results regarding the transformational leadership–follower creativity relationship (Banks *et al.*, 2016). That is, we show a sequential relationship that clarifies the relationship between transformational leadership and creative performance and aligns with research indicating that transformational leadership is positively related to creative performance (Qu *et al.*, 2015). Accordingly, we show that perspective taking and boundary spanning are the essential peripheral conditions that augment creative performance.

Second, we add to the creative performance literature and answer the calls for an empirical study of leadership influences on creativity (Hong *et al.*, 2018; Su *et al.*, 2019) by showing that leaders' cooperative exchange of accurate information with employees and the deliberate and systematic processing of such information affects knowledge workers' creative performance. Studies tend to focus on how transformational leadership encourages autonomy, self-concordance, self-efficacy and social identification with beneficiaries (Hoch *et al.*, 2018; Mittal and Dhar, 2015; Ng, 2017) but those mechanisms differ from prosocial influences on employees' perceptions of their relationships with work-teams (Goldstein *et al.*, 2014; Grant, 2012). We show that transformational leadership shapes creative performance by increasing relationships among supervisors, colleagues and external parties. Moreover, we validate creativity researchers who propose that transformational leaders use intrinsic motivational, psychological or cognitive processes to encourage creativity (Newman *et al.*, 2018). Our theory and empirical confirmation broaden the interactive margin of transformational leadership effects on creative performance among knowledge workers.

Our conceptual model extends prior understandings of motivated information processing theory by exclusively incorporating transformational leadership and creative performance. From the resource-based view (Barney, 1991), Chen *et al.* (2016) have found that both internal and external social capital of TMTs parallelly mediated the relationship between transformational leadership and organizational innovation. Our study extended Chen *et al.*'s (2016) study to the creation process of knowledge workers and used motivated information processing theory to explain why transformational leaders serve as an important source in incorporating information regarding others' preferences and priorities at the workplace. Transformational leaders are naturally driven to behave in an inspirational and charismatic manner deriving from their internalized perspective, thereby demonstrating openness, cooperation, trust and communication. Hence, transformational leaders are particularly instrumental in transmitting such cues into the work environment through their verbal and nonverbal behaviors regarding the importance of taking on colleagues and significant others' perspectives. Followers retrieve this information to understand how they are expected to behave in the workplace by looking beyond their internal teams to incorporate other viewpoints, explore unique options and champion new ideas. In a nutshell, this shapes their perceptions of perspective taking and boundary spanning. In doing so, the current study deepens our understanding of the integration of others' viewpoints by explaining how transformational leaders can foster the formation of a climate for perspective taking that enables all employees to feel comfortable applying their differences to work processes, tasks and strategies.

Perspective taking and boundary spanning have been separately and empirically linked to creative performance (Andersen and Kragh, 2015; Grant and Berry, 2011), but we clarify the positive sequential link between perspective taking and boundary spanning. Specifically, we build on theoretical arguments to show that perspective taking is linked to boundary spanning (Shih *et al.*, 2009). To the best of our knowledge, no previous study has assessed possible links between perspective taking and boundary spanning in a productive creative process. As creative employees also develop appropriate strategies for implementation (Gumusluoglu and Ilsev, 2009), our findings suggest that transformational leadership has important effects not only at the individual level especially among knowledge workers but also on innovation at the organizational level.

Practical implications

Our findings add to the literature on managerial practices, especially showing that transformational leadership promotes creative performance among knowledge workers (Fu and Liu, 2018; Gu *et al.*, 2018; Kim and Shin, 2017). Organizational training programs should encourage transformational leadership to increase employees' consideration of each other's interests, leading to increased perspective taking and boundary spanning for generating creative performance (Tippmann *et al.*, 2016). Thus, transformational leadership is recommended for directly and indirectly influencing creativity by encouraging information exchange (Andersen and Kragh, 2015), motivating knowledge workers and changing organizational attitudes. Thus, the leadership-motivation-behaviors model therefore shows that transformational leaders can transform information processes and encourage greater creativity.

Limitations and direction for future research

This study has some limitations. First, it is too narrowly focused on transformational leadership as the only antecedent. Future research should consider the four dimensions of transformational leadership and other leadership behaviors such as empowering leadership and transactional leadership (Afsar *et al.*, 2017; Zhang and Bartol, 2010). Second, we studied a single sample from China. Different samples would therefore enhance understandings of cross-level processes among creativity and leadership. China differs from other parts of the world in its viewpoints about motivational processes. For example, China has strong collectivist cultural values (Castañeda *et al.*, 2018; Li *et al.*, 2015) and high-power distance (Yuan and Zhou, 2015). Other contexts and cultural settings might confirm or challenge our conclusions. Third, our study only examined the sequential mediating roles of perspective taking and boundary spanning between transformational leadership and creative performance. As Liu *et al.* (2016) suggested, both prosocial motivation and epistemic motivation can encourage perspective taking. Future studies may investigate a complete chain mediation mechanism involving motivation (i.e. prosocial motivation and epistemic motivation), cognition (i.e. perspective taking) and behavior (i.e. boundary spanning) between transformational leadership and creative performance. Finally, creative performance could be measured with more objective indicators. Leadership's role in subordinate creative performance depends on creative outcomes (Herrmann and Felfe, 2014). Further studies on creative performance should use qualitative and quantitative methods to identify more significant correlations.

Conclusion

Notwithstanding its limitations, this paper brings a more precise interpretation of transformational leadership, perspective taking, boundary spanning and their influences on creative performance. Specifically, the findings by this research point out that enhancing

and promoting transformational leadership could be an outstanding asset to institutions that want to strengthen knowledge worker creative performance. Our study fills the existing research gaps in literature by discovering sequential mediating processes with other-focused motivational processes to understand creative performance as a new perspective. Similarly, future research must lay greater emphasis on the motivated information process as a mechanism for enhancing effective information sharing and creative performance.

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